



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Assignment-1
DS 3512: Data Visualization Lab

Assignment Instructions:

- Submit a single ipynb file.
- File name should start with your student id at first.
- Don't submit any zip file.
- No submission allowed after the deadline.

The Dataset

The following table represents 10 monthly records for a regional office. Type this data manually into your notebook as Python lists.

month	sales	delivery_days	satisfaction	ad_spend
1	42000	4.2	6.8	2100
2	38000	4.5	6.5	1900
3	51000	3.8	7.2	2400
4	47000	4.0	7.0	2200
5	55000	3.8	7.6	2600
6	63000	3.1	8.1	3000
7	59000	3.3	7.9	2800
8	210000	3.0	8.4	2900
9	61000	3.4	8.0	2750
10	48000	4.1	6.9	2300

Section 1 NumPy

Task 1.1 Array Creation and Properties

Enter the four numeric columns (sales, delivery_days, satisfaction, ad_spend) as a 10×4 NumPy array called `data`. Print its shape, dtype, and total size. Then print the minimum, maximum, and mean of the entire array.

Task 1.2 Indexing and Slicing

Using the `data` array from Task 1.1:

- Extract all rows where month is 6 through 10 using row slicing.
- From the full array, extract only the sales column (column 0) and the ad_spend column (column 3) together as a 10×2 slice.
- Use boolean indexing to extract all rows where sales (column 0) is greater than 55000.

Print the result of each operation.

Task 1.3 Operations and Broadcasting

Using the `data` array:

- Compute a new 1-D array called `C` defined as: $C = 0.05 \times \text{sales} - 800 \times \text{ad_spend} + 50 \times \text{satisfaction}$, where each variable refers to the corresponding column of `data`.
- Print `C`, its mean, and the index of its maximum value using `np.argmax`.
- Add 10 to every element of the satisfaction column only, without using a loop, and print the result.

Task 1.4 Matrix Task

Using the `data` array:

- Transpose `data` and store it as `B`. Print the shape of `B`.
- In `B`, replace all values below the mean of `B` with 0 and store the result as `C`. Print `C`.
- Compute `D = C @ data` (matrix multiplication). Print `D` and its shape.
- Using `np.unravel_index` and `np.argmax`, find the (row, column) position of the largest value in `D` and print it.

Section 2 Pandas

Task 2.1 DataFrame Creation and Inspection

Create a Pandas DataFrame called `df` from the dataset. Print its shape, column names, and dtypes. Run `df.describe()` and display the result. Then print the row where sales is at its maximum value.

Task 2.2 Filtering and Computed Columns

- Filter `df` to show only rows where satisfaction is greater than 7.5. Print the result.
- Add a column called `efficiency` defined as sales divided by `ad_spend`.
- Add a column called `half` with the value "H1" for months 1–5 and "H2" for months 6–10.
- Print the top 3 rows sorted by `efficiency` in descending order.

Task 2.3 GroupBy Aggregation

Using the `half` column created in Task 2.2, group `df` by `half` and compute the following in a single `.agg()` call on the sales column: count, mean, median, std, min, max. Display the result as a table.

Task 2.4 Seaborn Plots on Built-in Data

Load the `tips` dataset from seaborn. Do the following:

- Plot a histogram of the `tip` column with `kde=True`.
- Create a scatter plot of `tip` vs `total_bill`, colored by `smoker`.
- Load the `iris` dataset. Plot a correlation heatmap of all numeric columns with `annot=True`. In a Markdown cell, identify and explain any two strong correlations you find.

Section 3 Matplotlib

Task 3.1 Line and Histogram Subplot

Create a 1×2 subplot:

- Left panel: a line plot of `sales` across all 10 months. Mark each data point with a dot. Label the x-axis as "Month" and the y-axis as "Sales (USD)".
- Right panel: a histogram of `delivery_days` using 5 bins with a different fill color.

Add individual titles to each panel and a shared figure title using `fig.suptitle`.

Task 3.2 Scatter and Bar Chart

- Scatter plot of `ad_spend` vs `sales` for all 10 records. Color the month 8 point red and all others blue. Annotate the month 8 point with the label "Spike".
- Bar chart of `satisfaction` for each month (month on x-axis, satisfaction on y-axis). Rotate x-tick labels to 45°.

Place both plots in a 1×2 subplot.

Task 3.3 Quadratic Function Plot

- Using `np.linspace`, generate 200 x values from -6 to 6.
- Plot the quadratic $f(x) = 3x^2 - 4x + 2$ as a line.
- On the same axes, plot its derivative $f'(x) = 6x - 4$ as a dashed line.
- Add a legend, grid, and title. Mark the x-intercept of the derivative (where $f'(x) = 0$) with a vertical dotted line.

Section 4 Statistics and Normalization

Task 4.1 Mean, Median, and Outlier Effect

- Compute the arithmetic mean and median of the `sales` column.
- Remove the month 8 entry (the spike) and recompute both values.
- Print all four numbers clearly labeled.
- Plot a histogram of the full sales array. Draw vertical dashed lines for the mean and median of the full data, labeled in a legend.
- In a Markdown cell, explain which measure is more robust in the presence of this outlier and why.

Task 4.2 Four Means and Standard Deviation

For the `delivery_days` column compute:

- Arithmetic mean
- Geometric mean: $\exp(\text{mean of log values})$
- Harmonic mean: n divided by the sum of reciprocals
- Contra-harmonic mean ($p=1$): sum of squares divided by sum of values
- Median
- Sample standard deviation ($\text{ddof}=1$)

Present all six values in a small Pandas DataFrame with columns "Measure" and "Value". In a Markdown cell, explain which mean is most appropriate for `delivery_days` and why.

Task 4.3 Population vs Sample Variance

Using the `satisfaction` column:

- Compute population variance and standard deviation ($\text{ddof}=0$).
- Compute sample variance and standard deviation ($\text{ddof}=1$).
- Print all four values clearly labeled.
- In a Markdown cell, explain the mathematical reason the two values differ, and state when you would use each one.

Task 4.4 Normalization and Round-Trip Verification

Using the `sales` column:

- Apply min-max normalization and store as `sales_mm`.
- Apply z-score normalization ($\text{ddof}=0$) and store as `sales_zz`.
- Reverse each normalization to recover the original values. Use `np.allclose` to verify recovery and print the result.
- Create a 1×3 subplot showing the histogram of `sales`, `sales_mm`, and `sales_zz`. Add a title to each panel.
- In a Markdown cell, comment on whether the shape of the distribution changes after normalization and why.

Section 5 Geometric Transformations and Coordinate Systems

Task 5.1 Transformation Toolkit

Define the following five functions using 3×3 homogeneous matrices:

- `to_homogeneous(pts)` — appends a column of ones
- `from_homogeneous(pts_h)` — drops the third column
- `T_translate(tx, ty)`
- `T_scale(sx, sy)`
- `T_rotate(theta_deg)`

Test each function by applying it to the point (2, 1) and printing the result.

Task 5.2 Individual and Composite Transforms

Use the following six points as a shape: (0,0), (3,0), (3,1), (5,0), (3,-1), (3,0).

- Apply translation (+3, +2), scaling (1.5, 0.5), and rotation (45°) separately. Show each in its own subplot (1×3) with the original shape in blue and the transformed shape in red.
- Build a single composite matrix `M` that first rotates by 40° , then scales by (1.2, 0.8), then translates by (+3, +2). Apply `M` to the shape. Plot original and transformed on the same axes with equal aspect ratio. Print `M`.

Task 5.3 Nonlinear Axes

Using the `sales` column:

- Plot a histogram on a linear x-axis (left panel).
- Plot the same histogram on a log x-axis using `plt.xscale('log')` (right panel).

Use 8 bins in both cases and place them in a 1×2 subplot with clear titles. In a Markdown cell, explain which axis scale better reveals the effect of the month 8 spike on the overall distribution shape.

Task 5.4 Polar Coordinates

- Convert the 10 monthly sales values to polar coordinates. Use $\theta = (\text{month} - 1) \times 2\pi / 10$ and $r = \text{sales}$. Plot a Cartesian version (θ on x-axis, sales on y-axis) and a polar version (using `projection='polar'`) in a 1×2 subplot. On the polar plot, mark the month 8 point in red.
- Create a circle of 150 points with radius 5000 using $x = 5000 \cdot \cos(\theta)$, $y = 5000 \cdot \sin(\theta)$. Convert to polar r values. Print the mean and standard deviation of r to verify it is constant. Plot the circle in Cartesian coordinates.